

The pair singular series of a polynomial V : the dilated pieces below $H^{1/2}$

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Abstract

For a monic irreducible quadratic f without fixed prime divisor, part III decomposed the Cesàro off-diagonal $\text{Off}_f^*(H)$ into pieces indexed by the dilation u and proved that the pieces with $u > H^{1/2+\varepsilon}$ contribute a main term $c_{\text{off}}(f)H$. This paper treats dilations below that threshold. We prove that the pieces with $H^{0.47} < u \leq H^{0.53}$ contribute $O(H^{1-\delta})$, so that $\sum_{u > H^{0.47}} w(u) \mathcal{P}_u(H/u) = c_{\text{off}}(f)H + O(H^{1-\delta})$; the input is a bilinear estimate for Kloosterman fractions of Bourgain and Garaev together with a bound for the additive energy of reciprocals, valid for every modulus, which we deduce from an argument of Cilleruelo and Garaev. For negative discriminants we prove a second result at the other end: the pieces with $u \leq H^{1/3-c}$ contribute $O(H^{1-\delta(c)})$, $\delta(c) \approx 0.075c$, by running the automorphic kernel method of Grimmelt and Merikoski with the leading coefficient u^2 tracked. What remains is a neighbourhood of $u = H^{1/3}$.

1 Introduction

The problem. Let $f = t^2 + bt + c \in \mathbb{Z}[t]$ be monic, irreducible, without fixed prime divisor, of discriminant D . Part III [6] wrote the Cesàro off-diagonal of the pair singular series as a sum of *pieces*,

$$\text{Off}_f^*(H) = P(1) \sum_u \lambda(u) \omega(u) \mathcal{P}_u(H/u),$$

one for each squarefree *dilation* u , the piece $\mathcal{P}_u(Y)$ being the same problem at length $Y = H/u$ with the roots of $x^2 \equiv D$ replaced by those of $u^2 x^2 \equiv D$. Each piece is $O(Y)$, and the weights satisfy $\sum_{u \leq H} |w(u)|/u \asymp \log H$: the trivial bound for the whole sum is $H \log H$, which is exactly the size of the statement to be proved. Every piece must therefore be small, and the difficulty is uniformity in u .

Exponent coordinates. Write $u = H^\alpha$. Each method available for the pieces gives a saving $H^{-s(\alpha)}$ over the trivial bound on an interval of α , and degenerates at the ends of that interval. In these coordinates part III proves $s > 0$ for $\alpha > 2/3$ (averaging the dilations along u), and the present author's extension of that theorem, recorded in the revised part III, gives $s > 0$ for $\alpha > 1/2$ by a dispersion argument with Weil's bound. Below $1/2$ both mechanisms fail: the incomplete Kloosterman sums that dispersion produces have modulus $[d_1, d_2] \asymp Y^2$, and Weil's bound is trivial once the range of u is shorter than the square root of that modulus. This paper supplies two further mechanisms, one on each side of $\alpha = 1/3$.

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Results. Theorem 5 treats α near $1/2$: the weight w is decomposed as $1 * \chi_D * \gamma$, and the resulting bilinear forms in u are estimated by Weil's bound when one factor is long (Type I) or short (Type II), and, in the remaining balanced case, by the bilinear bound of Bourgain and Garaev [1] for Kloosterman fractions. That bound needs an estimate for the additive energy of reciprocals; the one in [1] is too weak at the sizes that occur here, and Lemma 2 supplies a stronger one, valid for every modulus, by transferring an argument of Cilleruelo and Garaev [2] for points on modular hyperbolas. Theorem 8 treats small α for negative discriminants: piece u is, after removing the content, a Type I sum for the polynomial $u^2\ell^2 + |D|$ in the sense of Grimmelt and Merikoski [3], and their automorphic kernel theorem [4] applies at level u^2e once the two kernel quantities are estimated with the dilation tracked. The diagonal is the number of orbits of the family, $\asymp u$; the off-diagonal, which their general bound would give as $\asymp u^2$, is in fact bounded independently of u (Lemma 6), because the distances inside the family are quantised in steps of $1/(4h)$ whatever the dilation.

What remains. Both mechanisms degenerate at $\alpha = 1/3$, from opposite sides and for unrelated reasons: there the balanced Type II atoms sit at $d^{1/4}$, where the energy input is not strong enough, and the kernel method's diagonal $\asymp u$ meets the trivial bound. The final section explains what the residual band costs. Two points are worth recording. A band of width 2δ in α costs $\delta \cdot H(\log H)^{1-c} \log \log H$, not $\delta H \log H$, because part III's Theorem E applies piece by piece inside it; so a failing method never makes the total worse. And if the savings on both sides vanish only at $\alpha = 1/3$, with implied constants that are powers of $\log H$, then the blocks near the point contribute $O(H)$ in total, which would give $\text{Off}_f^*(H) = O(H)$ — Hypothesis (E) of part I [5], a bounded second-order term in the Cesàro asymptotic. The constant A_f needs strictly more: a saving at the point itself.

2 Notation

We follow part III [6]. For a prime p let $\omega(p)$ be the number of roots of f modulo p , $\lambda(p) = p/(p - 2\omega(p))$, $P_p = 1 - \omega(p)^2/(p - \omega(p))^2$, $P(1) = \prod_p P_p$ and $w(u) = P(1)\lambda(u)\omega(u)$; w is supported on squarefree u and $\sum_{u \leq t} |w(u)| \ll t$. A modulus d' is *admissible* if it is squarefree, all its prime factors are split, and $(d', 2Du) = 1$; $R_{d'}^{(u)} = \{x \bmod d' : u^2x^2 \equiv D\}$, $\rho_k(d'; a) = \sum_{r \in R_{d'}} e(kar/d')$, and $\langle x \rangle_{d'} \in [1, d']$ is the representative of x . With $B_{d'}^{(Y)}(m) = \sum_{h \leq Y, h \equiv m \pmod{d'}} (Y - h) - Y^2/(2d')$, the piece is

$$\mathcal{P}_u(Y) = \sum_{d' > 1 \text{ admissible}} \frac{\lambda(d')}{d'} \sum_{x \in R_{d'}^{(u)}} B_{d'}^{(Y)}(\langle x \rangle_{d'}), \quad Y = H/u.$$

We write $G_{d'}(a, Y) = \sum_{r \in R_{d'}} B_{d'}^{(Y)}(\langle ar \rangle_{d'})$, $\bar{G}_{d'}(Y)$ for its mean over the units a , $G^\circ = G - \bar{G}$, and $\xi_k(d', N) = (1 - \cos(2\pi kN/d'))/\sin^2(\pi k/d')$ for the coefficients of the finite Fourier expansion of $B^{(N)}$ (part III, Lemma finfourier). Throughout, x^* denotes the inverse of x modulo the modulus at hand, $e_m(z) = e(z/m)$, and $\varepsilon > 0$ is arbitrarily small. Parameters: $\delta = 10^{-3}$, $\eta = \delta/10$, $\delta_3 = \delta$, $\varepsilon' = \eta/7$, $\kappa = \delta/2$, $K_1 = H^{2\eta}$.

3 The energy of reciprocals and bilinear Kloosterman fractions

We write x^* for the inverse of x modulo the modulus at hand and $e_m(z) = e(z/m)$.

Lemma 1 (smooth incomplete Kloosterman sums). *Let $q \geq 1$ be squarefree, $a \in \mathbb{Z}$, $X > 0$ and $T \geq 1$, and let W be smooth, supported in an interval of length at most $4X$, with $\|W\|_\infty \ll 1$*

and $\int |W''| \ll T^2/X$. Then

$$\sum_{(v,q)=1} W(v) e_q(av^*) \ll \tau(q) \left(1 + \frac{X}{q}\right) T^2 q^{1/2} (a, q)^{1/2}.$$

Proof. Split v into classes modulo q and apply Poisson summation in each: the sum equals $\sum_h \widehat{W}(h/q) S(a, h; q)/q$ with $S(a, h; q)$ the Kloosterman sum. For squarefree q , Weil's bound at the primes and twisted multiplicativity give $|S(a, h; q)| \leq \tau(q) (a, h, q)^{1/2} q^{1/2}$, which covers $h = 0$, where $S(a, 0; q) = c_q(a)$ is a Ramanujan sum and $|c_q(a)| \leq (a, q) \leq q^{1/2} (a, q)^{1/2}$. Since $\widehat{W}(\xi) \ll \min(X, T^2/(X\xi^2))$, one has $q^{-1} \sum_h |\widehat{W}(h/q)| \ll T^2(1 + X/q)$. $\square$

Lemma 2 (additive energy of reciprocals, every squarefree modulus). *Let $m \geq 2$ be squarefree, $1 \leq N \leq m$, and $J_4(N) = \#\{x \in [1, N]^4 : (x_i, m) = 1, x_1^* + x_2^* \equiv x_3^* + x_4^* \pmod{m}\}$. Then for every $\varepsilon > 0$,*

$$J_4(N) \ll_\varepsilon m^\varepsilon (N^2 + N^{7/2} m^{-1/2}).$$

Proof. Step 1: a hyperbola in a square box. Let $(\lambda, m) = 1$, $K \in \mathbb{Z}$ and $1 \leq M \leq m$; we claim $I := \#\{(x, y) \in (K, K+M]^2 : xy \equiv \lambda \pmod{m}\} \ll_\varepsilon m^\varepsilon (M^{3/2} m^{-1/2} + 1)$. For $M = 1$ this is trivial, so let $M \geq 2$. Writing $x = K + x'$, $y = K + y'$ with $1 \leq x', y' \leq M$, the congruence becomes $x'y' + Kx' + Ky' \equiv b := \lambda - K^2 \pmod{m}$. Put $T = \lfloor (m/M)^{1/2} \rfloor \geq 1$; by Dirichlet's theorem there are $1 \leq t \leq T$ and u_0 with $tK \equiv u_0 \pmod{m}$ and $|u_0| \leq \min(m/T, m/2)$. Multiplying by t ,

$$tx'y' + u_0x' + u_0y' = b_0 + zm, \quad |b_0| < m/2, \quad |z| \leq \frac{TM^2}{m} + \frac{2M}{T} + \frac{1}{2} \leq 5M^{3/2} m^{-1/2} + \frac{1}{2}.$$

For each z this is $(tx' + u_0)(ty' + u_0) = n_z := t(b_0 + zm) + u_0^2$, and $|n_z| \leq 7m^2$. Now $n_z \equiv t^2(x' + K)(y' + K) \equiv t^2\lambda \pmod{m}$, and λ is a unit with $1 \leq t^2 \leq m/M < m$, so $m \nmid n_z$ and in particular $n_z \neq 0$. Each z therefore contributes at most $\tau(|n_z|) \ll_\varepsilon m^\varepsilon$ pairs, and summing over the $\ll M^{3/2} m^{-1/2} + 1$ admissible z gives the claim.

Step 2: from energy to hyperbolas. Let $J_2(\lambda) = \#\{x, y \leq N, (xy, m) = 1 : x^* + y^* \equiv \lambda\}$, so $J_4 = \sum_\lambda J_2(\lambda)^2$ and $\sum_\lambda J_2(\lambda) = N'^2 \leq N^2$ with $N' = \#\{x \leq N : (x, m) = 1\}$. For a unit λ the congruence $x^* + y^* \equiv \lambda$ is equivalent to $(x - \lambda^*)(y - \lambda^*) \equiv \lambda^{*2}$, a unit hyperbola in a square box with equal shifts, so Step 1 gives $J_2(\lambda) \ll m^\varepsilon (N^{3/2} m^{-1/2} + 1)$. If $g = \gcd(\lambda, m) > 1$ then $p \mid g$ forces $x \equiv -y \pmod{p}$, hence $x \equiv -y \pmod{g}$ and $g \leq 2N$; modulo m/g the residue λ is a unit, so Step 1 applied there gives $J_2(\lambda) \ll m^\varepsilon (g^{1/2} N^{3/2} m^{-1/2} + 1)$ when $N \leq m/g$, while $J_2(\lambda) \leq N$ always. Using $\sum_{\gcd(\lambda, m)=g} J_2(\lambda) \leq N(2N/g + 1)$ and summing over the divisors $g \leq 2N$ of m yields the lemma; in the range $N > m/g$ one uses $4N^4/m \leq 4N^{7/2} m^{-1/2}$. $\square$

Lemma 3 (Bourgain–Garaev). *Let $m \geq 2$, $(a, m) = 1$, $1 \leq N_1, N_2 \leq m$, and let α_1, α_2 be coefficients bounded by 1 and supported on units. Then for all $k_1, k_2 \geq 1$,*

$$\left| \sum_{x_1 \leq N_1} \sum_{x_2 \leq N_2} \alpha_1(x_1) \alpha_2(x_2) e_m(ax_1^* x_2^*) \right|^{2k_1 k_2} \leq m N_1^{2k_1 k_2 - 2k_1} N_2^{2k_1 k_2 - 2k_2} J_{2k_1}(N_1) J_{2k_2}(N_2).$$

Proof. This is the inequality established in the proof of Theorem 3 of [1]: two applications of Hölder's inequality followed by Cauchy–Schwarz, using $\sum_\lambda J_k(\lambda) \leq N^k$ and that a is a unit. $\square$

Corollary 4. *For squarefree m and $m^{1/3} \leq N_1, N_2 \leq m$, under the hypotheses of Lemma 3, $|\sum \sum \alpha_1 \alpha_2 e_m(ax_1^* x_2^*)| \ll_\varepsilon m^\varepsilon (N_1 N_2)^{15/16}$.*

Proof. Take $k_1 = k_2 = 2$ in Lemma 3 and insert Lemma 2: for $N_i \geq m^{1/3}$ the second term of the energy bound dominates, $N_i^{7/2} m^{-1/2} \geq N_i^2$, so the right-hand side is $\ll m^{1+4\varepsilon} (N_1 N_2)^4 (N_1 N_2)^{7/2} m^{-1} = m^{4\varepsilon} (N_1 N_2)^{15/2}$. $\square$

4 The pieces around $u = H^{1/2}$

Theorem 5. *Let f be as above. There is $\delta_M > 0$ (any $\delta_M < 4\eta/7$ with the parameters above, so $\delta_M = 5 \cdot 10^{-5}$ is admissible) such that*

$$\sum_{H^{0.47} < u \leq H^{0.53}} w(u) \mathcal{P}_u(H/u) \ll_f H^{1-\delta_M}.$$

Consequently, with part III's Theorem on the range $u > H^{1/2+\varepsilon}$ applied at $\varepsilon = 0.03$, $\sum_{u > H^{0.47}} w(u) \mathcal{P}_u(H/u) = c_{\text{off}}(f)H + O(H^{1-\delta_M})$.

Proof of Theorem 5. The proof occupies the rest of this section. It is organised as a reduction to a Mellin-separated bilinear form (Step R1), a decomposition of the weight w with a covering of the resulting boxes (Step R5), and three estimates: a Type I bound (Step R2), a Type II bound (Step R3) and the residual balanced case (Step R4), the last being where Corollary 4 is used.

Notation for the proof. Put $\delta = 10^{-3}$, $\eta = \delta/10$, $\delta_3 = \delta$, $\varepsilon' = \eta/7$, $\kappa = \delta/2$ and $K_1 = H^{2\eta}$; all implied constants depend on f and on these parameters. For an admissible modulus d' and a unit a let $G_{d'}(a, Y) = \sum_{r \in R_{d'}} B_{d'}^{(Y)}(\langle ar \rangle_{d'})$, let $\bar{G}_{d'}(Y)$ be its mean over the units and $G^\circ = G - \bar{G}$; let $\rho_k(d'; a) = \sum_{r \in R_{d'}} e(kar/d')$, $\epsilon_{d'}(k) = c_{d'}(k)/\varphi(d')$ and $\xi_k(d', N) = (1 - \cos(2\pi kN/d'))/\sin^2(\pi k/d')$, and set $b_k(d') = \min(1/(4k^2), \pi^2(2Y)^2/(4d'^2))$.

Step R1: reduction to a Mellin-separated form. Partition the range of u smoothly into dyadic blocks, with ramps of relative width $H^{-\kappa}$ at the two endpoints; the transition zones contain $\ll H^{0.53-\kappa}$ values of u and contribute $\ll H^{1-\kappa}(\log H)^C$ by $\mathcal{P}_u(Y) \ll Y$. Fix a block, $U \in [H^{0.47}/4, 2H^{0.53}]$, $Y = H/U$, $a = \log U / \log H \in [0.46, 0.54]$. Then:

- (a) the dilation average contributes $\ll U(\log H)^C$, since $\sum_{d'} (\lambda(d')/d') |\bar{G}_{d'}(H/u)| \ll (\log Y)^C$ for every u ;
- (b) the moduli $d' > YH^\eta$ contribute $\ll H^{1-\eta+3\varepsilon'}$ by part III's Step 3, which is valid for every U ;
- (c) the moduli $d' \leq Y^{1-\delta}$ contribute $\ll UY^{1-\delta}$, by $|(\lambda/d')G^\circ| \leq |\lambda|\omega(1+\sigma/(8d'\varphi))$ and Shiu's theorem [7];
- (d) on the remaining range the finite Fourier expansion of part III applies at $N(u) = \lfloor H/u \rfloor$, the frequency $k = 0$ cancelling exactly because $R_{d'}$ is symmetric and $0 \notin R_{d'}$; the frequencies $k > K_1$ contribute $\ll H^{1-\eta}$, and replacing $\xi_k(d', N(u))$ by $\xi_k(d', H/u)$ costs $\ll U(\log H)^C$. Note $d' \geq Y^{1-\delta} > 2K_1$, so $(k, d') \leq K_1 < d'$;
- (e) with $F_{k,d'}(v) = \lambda(d')\xi_k(d', H/v)\phi(v/U)\psi(v)/(2d'^2)$ one has, for $T_0 = 1 + kY/d' + H^\kappa$ and $J \in \{0, 1, 2\}$, $\int |\tilde{F}_{k,d'}(it)|(1+|t|)^J dt \ll |\lambda(d')|(Y/(kd'))T_0^J \log H$, by the total-variation bound for $|t| \leq T_0$ and $J+2$ integrations by parts beyond it, using that the ramp's derivatives are supported on a set of relative measure $\ll H^{-\kappa}$; summing, $\sum_{k \leq K_1} \sum_{d'} |\lambda|\omega \int |\tilde{F}_{k,d'}| \ll Y(\log H)^3$.

The block therefore equals $(2\pi)^{-1} \int \sum_{k \leq K_1} \sum_{d'} \tilde{F}_{k,d'}(it) \mathcal{S}_{k,d'}(t) dt$ plus the errors above, where $\mathcal{S}_{k,d'}(t) = \sum_u w(u)u^{-it} \mathbf{1}_{(u,d')=1} [\rho_k(d'; u^*) - \omega(d')\epsilon_{d'}(k)]$, and u^{-it} factors over any factorisation of u . The mean terms contribute $\ll UY^\delta(\log H)^C$.

Step R5: decomposition and covering. Write $w = c_f(1 * \chi_D * \gamma)$ with χ_D the primitive character of the field, γ multiplicative with $\gamma(p) \ll 1/p$, $|\gamma(n)| \leq 20^{\omega(n)}$ and $\sum_n |\gamma(n)| n^{-1/2-\varepsilon} < \infty$ (for $\omega(2) = 1$ put $u = 2u'$ and absorb the Euler factor at 2 into γ). The part with $n_3 > H^{\delta_3}$ is $\ll UYH^{-\delta_3/2+\varepsilon}$. Decompose n_1, n_2, n_3 into smooth dyadic boxes N_1, N_2, N_3 , discarding boxes with $N_1 N_2 N_3 \not\asymp U$, which vanish. For $a \in [0.46, 0.54]$ the boxes split into three cases: either some atom has $N_i \geq Y^{1/2} H^{3\delta}$ (Step R2); or some atom lies in $W = [H^{3\delta}, UY^{-1/2} H^{-3\delta}]$ (Step R3); or both $N_1, N_2 \in [UY^{-1/2} H^{-3\delta}, Y^{1/2} H^{3\delta}]$ (Step R4). Indeed if $N_2 < H^{3\delta}$ then $N_1 \gg UH^{-3\delta-\delta_3} \geq Y^{1/2} H^{3\delta}$, because $U/Y^{1/2} \geq H^{0.19}$ on the block; the covering is exhaustive with margin 0.183.

Step R2 (Type I). Let an atom of length $N \geq Y^{1/2} H^{3\delta}$ carry the weight 1 (or χ_D restricted to a class modulo the conductor), the cofactor m' carrying arbitrary coefficients $\ll \tau_3(m') 20^\omega$. For fixed k, d', r, m', t , with $g = (k, d')$ and $m = d'/g$, the atom sum is $\sum_n W(n) e_m((k/g)r(nm')^*)$ with $W(n) = \phi(n/N) n^{-it}$ after removing $(n, g) = 1$ by Möbius, so Lemma 1 with $T = 1 + |t|$ applies and gives $\ll H^\varepsilon (1 + |t|)^2 (1 + N/m) m^{1/2}$. Summing over $m' \ll U/N$, the roots, the moduli against $\int |\tilde{F}|(1 + |t|)^2$ and $k \leq K_1$, and splitting the two branches of $T_0^2 \ll 1 + (kY/d')^2 + H^{2\kappa}$, the contribution is $\ll UY(H^{-3\delta+4\eta+1.5(1-a)\delta} + H^{-3\delta+\eta/2+2\kappa}) + UY^{1/2+O(\delta)} \ll UY H^{-1.79\delta}$.

Step R3 (Type II). Let $u = u_1 u_2$ with $|a(u_1)|, |b(u_2)| \ll H^\varepsilon$ and $U_2 \in W$. Apply Cauchy-Schwarz over (d', u_1) with weight $|\beta(d')|$, $\beta = \tilde{F}_{k,d'}(it)$, replacing the u_1 -range by a smooth majorant. The diagonal gives $S_1 \ll H^{2\varepsilon} U_1 \sum_{d'} |\beta|$. In S_2 the terms carrying $\epsilon_{d'}$ are negligible, and the main u_1 -sums are $\sum V(u_1/U_1) e_{d'}(\nu u_1^*)$ with $\nu \equiv k(ru_2^* - r'u_2'^*)$; Lemma 1 bounds each by $\tau(d')(1 + U_1/d') d'^{1/2} (\nu, d')^{1/2}$, and trivially by $3U_1$. Writing $(\nu, d') = (k, d')e$ with $e = \prod_{p|d', p \nmid k, ru_2' \equiv r'u_2(p)} p$, for fixed (d', r, r', u_2) there are at most $4U_2/e$ values of u_2' with that e when $e \leq 2U_2$, and at most one when $e > 2U_2$ — the latter case containing the diagonal and all degenerate pairs, since $d'/(k, d') > 2U_2$. Hence $S_2 \ll H^{O(\varepsilon+\eta)} \sum_{d'} |\beta| [U_2^2 Y^{1/2} + U_1 U_2^2 Y^{-1/2} + U_1 U_2]$, the middle term alone carrying the loss $Y^{\delta/2}$, and therefore the contribution is $\ll UY H^{-1.45\delta}$.

Step R4 (the residual). Here both atoms lie in $[UY^{-1/2} H^{-3\delta}, Y^{1/2} H^{3\delta}]$. For fixed k, d', r, n_3, t , with $g = (k, d')$ and $m = d'/g$ squarefree, $m \geq Y^{1-\delta} H^{-2\eta}$, the numerator $(k/g) r n_3^*$ is a unit modulo m and the double sum is of the shape treated in Corollary 4, with the coprimality conditions and the dyadic cut-offs absorbed into the coefficients. On the block, $\log N_i / \log m \geq 0.3462 > 1/3$ and $2N_i \leq m$, so Corollary 4 applies and gives $\ll m^\varepsilon (N_1 N_2)^{15/16}$. Since $N_1 N_2 \asymp U/n_3$, summing over n_3 against $|\gamma|$, over the roots, over the moduli against $\int |\tilde{F}|$ and over $k \leq K_1$ gives $\ll UY H^{-0.028}$.

Conclusion. Per block the errors are $H^{1-4\eta/7}$ from R1(b), $H^{1-\eta}$ from R1(d), $UY^{1-\delta}$ from R1(c), UY^δ from the mean terms, $H^{1-\delta_3/2+\varepsilon}$ from the n_3 -tail, $H^{1-\kappa}(\log H)^C$ from the ramps, and the three savings above. With $O(\log H)$ blocks the binding constraint is R1(b), and Theorem 5 follows for every $\delta_M < 4\eta/7$. $\square$

5 The small dilations

Throughout this section $D < 0$. We write $\Gamma_0(q)$ for the Hecke congruence subgroup, $\Gamma_0^0(u) = \{\gamma : u \mid b, u \mid c\}$, and for positive definite forms $g_i = \begin{pmatrix} A_i & B_i \\ B_i & C_i \end{pmatrix}$ we put $\langle g_1, g_2 \rangle = A_1 C_2 + A_2 C_1 - 2B_1 B_2$, so that the point-pair invariant of the associated points of the upper half plane is $u(g_1, g_2) = (\cosh d - 1)/2$ with $\cosh d = \langle g_1, g_2 \rangle / (2\sqrt{\det g_1 \det g_2})$. $\mathcal{S}_{ah}$ denotes the set of all positive definite forms of determinant ah ; the family $\mathcal{S}_{a,h}(q_0)$ (with a comma) is defined below. The kernel method

of Grimmelt and Merikoski is used in the form of their Theorem 8.1 [4] and Lemma 3.1 [3], in their notation $(\Delta_q, K_q, \text{the kernel quantities } \mathcal{K}_1(q; Z, R), \mathcal{K}_2(q; Z), \text{the functional } \alpha_q \text{ on the Heegner set})$. The parameters of Theorem 8 are δ, η, η' (unrelated to the parameters of Theorem 5), fixed in Step S8; $\theta = 7/64$ is the Kim–Sarnak bound, valid uniformly for congruence subgroups.

5.1 The family of a dilated piece, and its normal form

The piece $\mathcal{P}_u(Y)$ counts the roots of $Q_u(\ell) = u^2\ell^2 + |D|$ modulo admissible k , which are coprime to $2Du$. The content of Q_u is $c = \gcd(u^2, |D|)$, and since u is squarefree, $v_p(c) = \min(2, v_p(D))$ for $p \mid u$, so $c = Ps^2$ with

$$P = \prod_{p \mid u, v_p(D)=1} p, \quad s = \prod_{p \mid u, v_p(D) \geq 2} p, \quad u' = \frac{u}{Ps}, \quad a = Pu'^2, \quad h = \frac{|D|}{Ps^2}, \quad h_0 = Ph = \frac{|D|}{s^2}.$$

Then $\gcd(a, h) = 1$, $P \parallel h_0$, $h_0 = O_D(1)$, $(u', h_0) = 1$ and $a/h = u^2/|D|$; moreover u' is odd: either D is odd and w is supported on odd u (a root of f modulo 2 would be a fixed prime divisor), or D is even, hence $4 \mid D$, and w is supported on even u with the prime 2 entering s (part III, §2). Since the primes of c divide u and k is coprime to u , $k \mid Q_u(\ell)$ if and only if $k \mid a\ell^2 + h$. The piece is thus a Type I sum for $a\ell^2 + h$ in the sense of [3]: by Lemma 3.1 there, the pairs (k, ℓ) with $k \equiv 0 \pmod{q_0}$ and $k \mid a\ell^2 + h$ correspond to the family

$$\mathcal{S}_{a,h}(q_0) = \left\{ g = \begin{pmatrix} A & B \\ B & C \end{pmatrix} : AC - B^2 = ah, A, C > 0, a \mid B, aq_0 \mid C \right\}$$

at level $q = aq_0$, the functional being $\alpha_q(g) = \mathbf{1}_{q \mid c(g)} \mathbf{1}_{a \mid b(g)}$ on the Heegner set Λ_{ah} (their “ $b^2 - ac = ah$ ” is a slip for $ac - b^2 = ah$). Two facts about this family are used repeatedly. First, $\mathcal{S}_{a,h}(q_0)$ is *empty* unless $\gcd(a, q_0) = 1$: for $p \mid P$, $aq_0 \mid C$ and $a \mid B$ give $p^2 \mid AC - B^2 = ah$ against $v_p(ah) = 1$, and for $p \mid u'$ they give $p^3 \mid C$, $p^4 \mid B^2$ against $v_p(ah) = 2$. Second, $a \mid \langle g_1, g_2 \rangle$ termwise, so $N = \langle g_1, g_2 \rangle / u'^2$ is an integer. In our earlier drafts the case $P = s = 1$, $a = u^2$, was treated alone; the general normal form costs nothing extra.

5.2 The off-diagonal at one level does not grow with the dilation

Lemma 6. *Let (a, h) be as in §5.1, $F = \mathcal{S}_{a,h}(1)$, and for $Z \geq 1$*

$$\text{off}(a; Z) = \sum_{g_2 \in \Gamma_0(a) \setminus F} \sum_{\substack{g_1 \in F, g_1 \neq g_2 \\ u(g_1, g_2) \leq Z}} (1 + u(g_1, g_2))^{-1/2},$$

the orbits counted with the $\text{SL}_2(\mathbb{Z})$ -stabiliser weights $|\Gamma_g|^{-1} \leq 1$. Then $\text{off}(a; Z) \ll_{D,\varepsilon} (Zu)^\varepsilon (1 + Z^{1/2})$, uniformly in u : the u -dependence is the divisor factor $6^{\omega(u')}$, absorbed by $(Zu)^\varepsilon$, and there is no power of h .

Proof. (1) *Undilation.* Let $\iota = \text{diag}(1, 1/u')$ and $G = \iota g \iota$. For $g \in F$ write $B = aB_1 = Pu'^2 B_1$ and $C = aC_1 = Pu'^2 C_1$; then $G = \begin{pmatrix} A & Pu' B_1 \\ Pu' B_1 & PC_1 \end{pmatrix}$ has $\det G = ah/u'^2 = h_0$, and $g \mapsto G$ is a bijection of F onto $F' = \{G : \det G = h_0, Pu' \mid B_G, P \mid C_G\}$, with $\iota \Gamma_0(a) \iota^{-1} = \{\gamma : u' \mid b, Pu' \mid c\} = \Gamma_0(P) \cap \Gamma_0^0(u')$ (using $(P, u') = 1$). The map is an isometry of the associated points, and $\langle g_1, g_2 \rangle = u'^2 \langle G_1, G_2 \rangle = u'^2 N$ with $1 + 2u(g_1, g_2) = N/(2h_0)$. Hence $N > 2h_0$ for $g_1 \neq g_2$, and

$$\text{off}(a; Z) = \sum_{2h_0 < N \leq 2h_0(1+2Z)} \left(\frac{4h_0}{N + 2h_0} \right)^{1/2} P_u(N), \quad P_u(N) = \#\{(\Gamma_0(P) \cap \Gamma_0^0(u'))\text{-orbits of pairs } (G_1, G_2) \in F\}$$

(2) *Lift to level one.* The stabiliser in $\text{SL}_2(\mathbb{Z})$ of a pair of distinct points is $\pm I$, so each orbit lies in a unique $\text{SL}_2(\mathbb{Z})$ -orbit of pairs of determinant- h_0 forms with $\langle G_1, G_2 \rangle = N$, and $P_u(N) =$

$\sum_{\mathcal{O}} \#\{\tau \in (\Gamma_0(P) \cap \Gamma_0^0(u')) \backslash \mathrm{SL}_2(\mathbb{Z}) : \tau.G_1, \tau.G_2 \in F'\}$ over the $P_1(N)$ such orbits, evaluated at any representative (the count does not depend on it). Taking G_2 among the finitely many reduced forms of determinant h_0 and observing that $W = 2h_0G_1 - NG_2$ lies in the rank-two lattice $G_2^\perp$ with $-\det W = h_0(N^2 - 4h_0^2)$, the number of representations of an integer n by a fixed definite binary lattice, $\ll_\varepsilon n^\varepsilon$, gives $P_1(N) \ll_{D,\varepsilon} N^\varepsilon$; $h_0 = O_D(1)$ is what makes this uniform. (3) *The local count.* By the Chinese remainder theorem the coset space factors over the primes of Pu' ; parametrise $T_p \backslash \mathrm{SL}_2(\mathbb{F}_p)$ by the ordered pair of distinct lines spanned by the rows of τ , and $\Gamma_0(p) \backslash \mathrm{SL}_2(\mathbb{F}_p)$ by the second row $L_2 \in \mathbb{P}^1(\mathbb{F}_p)$. Then $p \mid B(\tau.G)$ reads $L_1 \perp_G L_2$ and $p \mid C(\tau.G)$ reads $Q_G(L_2) = 0$. For $p \mid u'$ (so $p \nmid 2h_0$) the condition is $u' \mid B$ only: for one form there are $p - \chi_{-h_0}(p)$ cosets; for a pair, L_2 must be an eigenline of $G_2^{-1}G_1$, whose characteristic polynomial is $\lambda^2 - (N/h_0)\lambda + 1$, so there are at most 2 cosets unless $G_2^{-1}G_1$ is scalar modulo p , in which case $p^2 \mid N^2 - 4h_0^2$ and there are at most $p + 1$. For $p \mid P$ both conditions are imposed, and they collapse to one: $p \mid B_G$, $p \mid C_G$ and $p \parallel h_0 = \det G$ force $p \nmid A_G$, so $G \equiv \mathrm{diag}(A_G, 0)$ is of rank one modulo p , its unique projective zero is its kernel line L_0 , and for $L_2 = L_0$ one has $GL_2 \equiv 0$, so $L_1 \perp_G L_2$ for every L_1 . Hence the local factor at $p \mid P$ is 1, for one form and for pairs, and altogether the number of cosets for a pair is at most $2^{\omega(u')}g_2(N)$ with $g_2(N) = \prod_{p \mid u', p^2 \mid N^2 - 4h_0^2} p$. (4) *The sum over N .* With $M = 2h_0(1 + 2Z)$ and d running over the divisors of u' with $d \leq M$ (which is automatic, since $d^2 \mid N^2 - 4h_0^2 < M^2$),

$$\sum_{2h_0 < N \leq M} N^{-1/2} g_2(N) \leq \sum_{d \mid u', d \leq M} d \sum_{\substack{N \leq M \\ d^2 \mid N^2 - 4h_0^2}} N^{-1/2} \leq \sum_{d \mid u', d \leq M} d \cdot 2^{\omega(d)} (d^{-1/2} + 2M^{1/2}/d^2) \leq 3\tau_3(u')M^{1/2},$$

because the $2^{\omega(d)}$ classes $N \bmod d^2$ with $d^2 \mid N^2 - 4h_0^2$ each have $N \geq d$. Combining (1)–(4), $\mathrm{off}(a; Z) \ll_{D,\varepsilon} M^\varepsilon 2^{\omega(u')} \tau_3(u') M^{1/2} \ll (Zu)^\varepsilon (1 + Z^{1/2})$. $\square$

The mechanism is that the distances between family points are quantised: $\cosh d \in (2h_0)^{-1}\mathbb{Z}$ independently of the dilation, so the number of pairs at distance $\leq Z$ is governed by the divisors of $N^2 - 4h_0^2$, not by the size of the level. Numerically, for $a = 3u'^2$ and $a = 15u'^2$ ($D = -3, -15$) the diagonal is exactly $c \prod_{p \mid u'} (p - \chi_{-h_0}(p))$ with the factor at the primes of P equal to 1, and $\mathrm{off}(a; Z)/Z^{1/2}$ shows no growth in u' ; the local count of step (3) was also checked by enumerating $\mathrm{SL}_2(\mathbb{F}_p)$ for 141 normal forms, nine of them with D non-fundamental.

Remark 7 (on the implied constant imported from the kernel theorem). Theorem 8.1 of [4] is stated with an unspecified factor $\delta_{\mathrm{GM}}^{-O(1)}$, where δ_{GM} is the smoothness parameter of the input function. Following its proof through §7.1–§7.4 — the unskewing with $R_1 = X$, $R_2 = 1$ is loss-free; F of right type 0 forces $\ell_2 = 0$, so that the second K -type sum is absent and the α_2 -side majorant may be taken at $L = 1$; and $\delta_1 = \delta$ because $X \geq Y$, which here reads $Lu \geq |D|^{1/2}$ and holds on every non-trivial box — one may take $\delta_{\mathrm{GM}}^{-6}$. We do not optimise this and use only that the exponent is absolute: we fix $C = 6$, cite it as “following the proof one may take”, and choose the parameters *after* C ; the shape $\delta_S(c) \asymp c$ of Theorem 8 does not depend on the value of C , only the constant does.

5.3 Theorem S

Theorem 8. *Let f be as above with $D < 0$ (any discriminant, fundamental or not). For every $c > 0$,*

$$\sum_{u \leq H^{1/3-c}} w(u) \mathcal{P}_u(H/u) \ll_{f,c} H^{1-\delta_S(c)}, \quad \delta_S(c) = \frac{3(1-2\theta)}{32-6\theta} c \approx 0.0748c,$$

with $\theta = 7/64$; the parameters below are $\delta = \eta = \eta' = t$, $t = \frac{9(1-2\theta)}{2(32-6\theta)} c \approx 0.112c$, and $\delta_S = 2t/3$.

The proof is written for the general normal form of §5.1. The primes of (u, D) and the prime 2 need no hypothesis, and none could be imposed: for even D the weight w is supported on even u , so a restriction $(u, D) = 1$ would make the theorem empty there.

Step S1: the reduction

Fix u squarefree in the support of w with $u \leq H^{1/3-c}$, $Y = H/u$, and let (a, h) be its normal form. Define, for every $k \geq 1$, $\varrho(k) = \#\{x \bmod k : ax^2 + h \equiv 0\}$; for admissible k this is $|R_k^{(u)}|$, and it is what the kernel method sees at every level, whereas $|R_k^{(u)}|$ counts p roots of $u^2x^2 \equiv D \equiv 0$ at a prime $p \mid P$ where $\varrho(p) = 0$. Since $R_k^{(u)}$ is symmetric and $0 \notin R_k^{(u)}$ (as k is coprime to $2D$), pairing $x \leftrightarrow k - x$ in the definition of the piece gives the exact identity, for admissible k ,

$$\sum_{x \in R_k^{(u)}} B_k^{(Y)}(\langle x \rangle_k) = \frac{1}{2} \sum_{\ell \neq 0, k \mid a\ell^2 + h} T(\ell) - \frac{\varrho(k)}{2k} \int_{\mathbb{R}} T, \quad T(t) = (Y - |t|)^+, \quad \int_{\mathbb{R}} T = Y^2, \quad (1)$$

the shape “count minus density times mass” of [3]. The small moduli $k \leq Y^{1-\delta}$ contribute $\ll \sum_{k \leq Y^{1-\delta}} |\lambda(k)| \varrho(k) \ll Y^{1-\delta} (\log Y)^{O(1)}$, since for each pair $\{x, k - x\}$ the bracket $B_k^{(Y)}(m) + B_k^{(Y)}(k - m)$ lies in $[-k/4, k/4]$ (part I, proof of Theorem 5) and Shiu’s theorem bounds the sum; no smoothing is needed there, the identity being exact for each k . The large moduli $k > Y^{1+\eta}$ contribute $\ll Y^{1-\eta} (\log Y)^{11}$ by part III’s Lemma tail, uniformly in $u \leq Y^3$. To hand a smooth weight to the kernel theorem we insert ψ_1 , supported on $[Y^{1-\delta}/2, 2Y^{1+\eta}]$ and equal to 1 on $[Y^{1-\delta}, Y^{1+\eta}]$, and remove the two transition ranges by the same two bounds. On the window we write, in this order,

$$\lambda(k) \mu^2(k) \mathbf{1}_{(k, 2D)=1} = \sum_{j \mid (k, \text{rad } 2D)} \mu(j) \sum_{e \mid k} \kappa(e) \sum_{\substack{m^2 \mid k \\ (m, 2D)=1}} \mu(m),$$

with κ supported on split squarefree integers (so $(e, 2D) = 1$ automatically, and $1 * \kappa$ reproduces λ only after the j -Möbius) and $(k, u) = 1$ automatic (a prime of (k, u) would divide $|D|$, hence $2D$). The restriction $(m, 2D) = 1$ is legitimate because the inner (e, m) -sum does not depend on j , so $\sum_{j \mid (k, \text{rad } 2D)} \mu(j) = \mathbf{1}_{(k, 2D)=1}$ annihilates every k with $(k, 2D) > 1$ whatever the inner sum is, and on $(k, 2D) = 1$ the restricted inner sum is still $\lambda \mu^2(k)$. The tail $m > M := Y^\eta$ is removed *before* any further decomposition at a cost $\ll Y^{1-\eta+\varepsilon}$, and no truncation in e is needed. Then T is decomposed into smooth dyadic boxes $T_L(\ell) = Y \psi_2(\ell/L)$ in $|\ell| \sim L$, the two kinks of T at $\ell = \pm Y$ smoothed at scale $\Delta = Y^{1-\eta'}$, which changes the sum by $\ll Y^{1+\delta-2\eta'+\varepsilon}$ in total and gives $\psi_2^{(J)} \ll Y^{J\eta'}$. A box (L, K) , K dyadic in $[Y^{1-\delta}, Y^{1+\eta}]$, has trivial size $\ll (Y/K) L H^\varepsilon$, so the boxes with $L \leq Y^{1-3\delta}$ contribute $\ll Y^{1-2\delta+\varepsilon}$ together and the kink of T at $\ell = 0$ never matters. The outcome: for every u ,

$$\mathcal{P}_u(Y) = \sum_{\text{boxes } (L, K)} \sum_{j, e, m \leq M} \mu(j) \kappa(e) \mu(m) \mathfrak{B}(L, K; q_0) + O(Y^{1-\min(\delta, \eta, 2\eta'-\delta)+\varepsilon}), \quad q_0 = \text{lcm}(j, e, m^2), \quad (2)$$

with $O(\log^2 H)$ boxes, $Y^{1-3\delta} \leq L \leq Y$, and $\mathfrak{B}(L, K; q_0) = \frac{1}{2} \sum_{k \equiv 0 \pmod{q_0}} \frac{1}{k} \psi_1(k/K) [\sum_{k \mid a\ell^2 + h} T_L(\ell) - \varrho(k) \int T_L/k]$. The levels are $q = aq_0$; at the levels with $\gcd(a, q_0) > 1$ both terms of $\mathfrak{B}$ vanish identically — the family is empty by §5.1, and $\varrho(k) = 0$ for $k \equiv 0 \pmod{q_0}$ by the same valuation count — so only the levels with $\gcd(a, q_0) = 1$ remain, and there Lemma 3.1 of [3] applies in its stated form $q = a \cdot q_0$.

Step S2: the two-scale form

Following §5 of [3], replace T_L by a second weight $T^{(2)}(\ell) = (\int T_L) \psi(\ell/X_2)/(X_2 \int \psi)$ of the same mass at the scale $X_2 = K^{1+2\eta}$, which exceeds $L \leq Y$ since $(1-\delta)(1+2\eta) > 1$ for $\delta = \eta$ small. The density terms of the two scales are identical, so $\mathfrak{B}(L, K; q_0) = \mathcal{A}_1 - \mathcal{A}_2 + \mathcal{E}$, where $\mathcal{A}_i = \langle I | \Delta_q F_i | \alpha_q \rangle$ is built from (ψ_1, T_L) , resp. $(\psi_1, T^{(2)})$, exactly as in their §5, and $\mathcal{E}$ is the second-scale sum $\sum_{k \equiv 0(q_0)} \frac{1}{k} \psi_1(k/K) [\sum_{k|a\ell^2+h} T^{(2)}(\ell) - \varrho(k) \int T^{(2)}/k]$, which Poisson summation modulo k evaluates: for $k \sim K$ the range X_2 exceeds k by the factor $K^{2\eta}$, the dual frequencies are $\geq K^{2\eta}$, and $\mathcal{E} \ll_A K^{-A}$ for every A , uniformly in u (constants depending on A, η, ψ). The skew of F_1 is $X/Y = L\sqrt{a/h} = Lu|D|^{-1/2}$ with x -range $R = L/K \in [Y^{-\eta-3\delta}, Y^\delta]$; that of F_2 is larger by X_2/L , and its bound is smaller than that of $\mathcal{A}_1$ by $(L/X_2)^{1/2-\theta}$, so only $\mathcal{A}_1$ is estimated below.

Step S3: the hypotheses of the kernel theorem

In Iwasawa coordinates $x = \ell/k$, $y = \sqrt{h/a}/k = \sqrt{|D|}/(uk)$ the weight of $\mathcal{A}_1$ is $F(g) = (Y/K)f(x, y)$ with $f(x, y) = \psi_2(\ell/L)\tilde{\psi}_1(k/K)$, $\tilde{\psi}_1(t) = \psi_1(t)/t$; f is supported on $|x| \asymp L/K$, $y \asymp \sqrt{|D|}/(uK)$, is not a product (the argument of ψ_2 depends on y too), and each ∂_y produces a factor $1/y$, so $\partial_x^{J_1} \partial_y^{J_2} f \ll (Y^{\eta'})^{J_1+J_2} (L/K)^{-J_1} (\sqrt{|D|}/(uK))^{-J_2}$, the only non-dyadic scale being the kink smoothing. After splitting the x -support (of ratio ≤ 4) into $O(1)$ smooth dyadic pieces, $f \in C_{\delta_{\text{GM}}}^{10}(X, Y)$ in the sense of Definition 2 of [4] with $\delta_{\text{GM}}^{-1} \asymp Y^{\eta'}$, and $X/Y = Lu|D|^{-1/2} > \delta_{\text{GM}}$ on every box. The Hecke variable is trivial, so their Rankin–Selberg input is not invoked; $q^{o(1)} = (aq_0)^{o(1)} = H^{o(1)}$; and the unspecified $\delta_{\text{GM}}^{-O(1)}$ of Theorem 8.1 is δ_{GM}^{-C} with $C = 6$ by Remark 7. Theorem 8.1 then gives, for every $Z_0, Z_1, Z_2 \geq 1$ with $Z_0 Z_1 Z_2 \geq X/Y + 1$,

$$|\mathcal{A}_1| \ll H^{o(1)} Y^{C\eta'} \frac{Y}{K} (Lu)^{1/2} Z_0^\theta \mathcal{K}_1(q; Z_1, R)^{1/2} \mathcal{K}_2(q; Z_2)^{1/2} \quad (3)$$

(with $(X/Y)^{1/2} \leq (Lu)^{1/2}$).

Step S4: the first kernel quantity

The example in §1.2 of [4] gives, for $q \geq 1$, $Z \geq 1$ and $0 < R \leq H^{O(1)}$, $\mathcal{K}_1(q; Z, R) \ll H^{o(1)}(1 + R + 1/(qR) + Z/q)$ (the cases $c = 0$, $b = 0$, $bc \neq 0$ of $\gamma \in \Gamma_0(q)$; we checked it numerically at our levels). Summed over the levels $q = a \text{ lcm}(j, e, m^2)$ of a block $e \sim E$, $m \sim M$, with $R \leq Y^\delta$ and $1/R \leq Y^{\eta+3\delta}$,

$$\sum_{\text{levels}} \mathcal{K}_1 \ll_D H^{o(1)} (Y^\delta EM + Y^{\eta+3\delta} + Z_1/a) \leq H^{o(1)} Y^{\eta+3\delta} (EM + Z_1/a). \quad (4)$$

Step S5: the second kernel quantity, the dictionary and the diagonal

By definition (§4.1 of [3]) and since $\alpha_q \geq 0$ and the subtracted volume term is ≥ 0 , $\langle \alpha_q | \Delta_q k | \alpha_q \rangle \leq \langle \alpha_q | K_q k | \alpha_q \rangle$, and the smooth kernel may be replaced by the majorant $\mathbf{1}_{u \leq Z_2^2} (1+u)^{-1/2}$. Unfolding the γ -sum once gives the orbit form $\text{diag}(q) + \text{off}(q; Z_2)$ with the $\text{SL}_2(\mathbb{Z})$ -stabiliser weights on the g_2 -side and the g_1 -side unweighted (with $\Gamma_0(q)$ -stabilisers the identity fails by a factor 2). Unfolding a second time, as in Proposition 4.1 of [3],

$$\text{off}(aq_0; Z_2) = \sum_{z_2 \in \Lambda_{ah}} |\Gamma_{z_2}|^{-1} \sum_{\substack{w_1 \in \mathcal{S}_{ah} \\ w_1 \neq z_2}} k(u(w_1, z_2)) n_{aq_0}(w_1, z_2), \quad n_q(w_1, z_2) = \#\{\tau \in \Gamma_0(q) \backslash \text{SL}_2(\mathbb{Z}) : \alpha_q(\tau \diamond w_1) = z_2\} \quad (5)$$

and for $\gcd(a, q_0) = 1$ the Chinese remainder theorem gives $n_{aq_0} = n_a n_{q_0}$ with $n_{q_0} = \#\{(c_0 : d_0) \in \mathbb{P}^1(\mathbb{Z}/q_0) : q_0 \mid Q_{w_1}(c_0, d_0), q_0 \mid Q_{z_2}(c_0, d_0)\}$. Summing n_a over the $\text{SL}_2(\mathbb{Z})$ -orbits of pairs

with invariant N is the count $P_u(N)$ of Lemma 6: the a -part of the off-diagonal is that lemma, and the q_0 -part is the level sum below. The diagonal is the weighted number of $\Gamma_0(q)$ -orbits of the family, $\text{diag}(q) = c_a \rho^*(q_0)$ with $c_a \asymp_D u$ (exactly $\prod_{p|u'} (p - \chi_{-h_0}(p))$ times a factor $O_D(1)$, the primes of P contributing 1) and $\rho^*(q_0) \leq 2^{\omega(q_0)} \prod_{p|(q_0, 2D)} (p+1) \ll_D 2^{\omega(q_0)}$ the number of projective zeros modulo q_0 ; hence $\sum_{\text{levels}} \text{diag} \ll_D H^{o(1)} uEM$.

Step S6: the level-summed off-diagonal

For $1 \leq Z_2 \leq H^{O(1)}$ we claim

$$\sum_{\text{levels } q_0 \text{ of a block}} \text{off}(aq_0; Z_2) \ll_{f, \varepsilon} H^\varepsilon (1 + Z_2), \quad (6)$$

uniformly in u ; the levels are those with $\gcd(a, q_0) = 1$, and they may have $(q_0, s) > 1$. Fix a pair (w_1, z_2) of distinct family points with invariant $N = \langle w_1, z_2 \rangle / u'^2 \in (2h_0, M]$, $M = 2h_0(1 + 2Z_2^2)$. A common projective zero of Q_{w_1}, Q_{z_2} modulo p^k forces $p^k \mid \text{Res} = \langle w_1, z_2 \rangle^2 - 4 \det w_1 \det z_2 = u'^4(N^2 - 4h_0^2)$, because $x^3 \text{Res}, y^3 \text{Res} \in (Q_{w_1}, Q_{z_2})$, and Hensel's lemma gives at most 2 zeros of each form modulo p^k when $p \nmid 2ah$. The $O_D(1)$ primes $p \mid 2D$ are pulled out first with the trivial factor $p+1$ each, at most once since $v_p(q_0) \leq 1$ there (j is squarefree and $(em, 2D) = 1$); the primes of u' do not divide q_0 ($\gcd(a, q_0) = 1$), so the u'^4 is inert; and every remaining prime satisfies $p \nmid 2Dah$, since $a = Pu'^2$ and $h \mid D$. The number of (j, e, m) with a given lcm is at most $\tau(q_0)^2 = H^{o(1)}$, so

$$\sum_{(j, e, m): e \leq E, m \leq M} n_{\text{lcm}(j, e, m)}(w_1, z_2) \leq \prod_{p \mid 2D} (p+1) \cdot H^{o(1)} \sum_{q'_0 \mid N^2 - 4h_0^2, (q'_0, 2Dah)=1} 2^{\omega(q'_0)} \ll_D H^{o(1)} \tau_3(N^2 - 4h_0^2),$$

using $\sum_{d|n} 2^{\omega(d)} \leq \tau_3(n)$: the level sum costs a divisor factor, not a factor EM^2 . Now $N^2 - 4h_0^2 \geq 4h_0 + 1 > 0$ for distinct points, $\tau_3(N^2 - 4h_0^2) \ll M^\varepsilon$, $M \asymp_D Z_2^2 \leq H^{O(1)}$, and Lemma 6 (steps (2)–(4): the weight $\ll_D N^{-1/2}$, $P_1(N) \ll_D M^\varepsilon$, and $\sum_{2h_0 < N \leq M} N^{-1/2} g_2(N) \leq 3\tau_3(u') M^{1/2}$) give $\sum_{q_0} \text{off}(aq_0; Z_2) \ll_{D, \varepsilon} H^\varepsilon 2^{\omega(u')} \tau_3(u') M^{1/2} \ll H^\varepsilon (1 + Z_2)$, the u -factor $6^{\omega(u')}$ being pointwise a divisor-size factor and averaged in the application by $\sum_{u \sim U} 6^{\omega(u)} / u \asymp (\log U)^5$. This proves (6). (The hypothesis $Z_2 \leq H^{O(1)}$ is where the ε -powers, all of which are M^ε , become H^ε ; it holds in Step S7 after the dichotomy there.) Together with Step S5, $\sum_{\text{levels}} \mathcal{K}_2 \ll H^{o(1)} (uEM + Z_2)$.

Step S7: the per-box bound

Fix a box (L, K) and a block (E, M) of levels, and apply (3) to every level with the same Z_0, Z_1, Z_2 . Pull the coefficients out by $|\kappa(e)\mu(m)| \ll H^{o(1)}/E$ — μ has no decay in m , and this costs a factor M — and apply Cauchy–Schwarz over the EM levels:

$$\sum_{\text{levels}} |\kappa(e)\mu(m)| |\mathcal{A}_1| \ll H^{o(1)} Y^{C\eta'} \frac{Y}{K} \frac{1}{E} (Lu)^{1/2} Z_0^\theta \left(\sum \mathcal{K}_1 \right)^{1/2} \left(\sum \mathcal{K}_2 \right)^{1/2}.$$

If $uEM \leq L^{1/2}$ choose $Z_1 = u^2 EM$ (so $Z_1/a = Ps^2 EM = O_D(1) EM$), $Z_2 = uEM$ and $Z_0 = 2L/(u^2 E^2 M^2) \geq 2$; then $Z_0 Z_1 Z_2 = 2Lu \geq X/Y + 1$, and by (4) and Step S6 the right-hand side is $\ll H^{o(1)} Y^{C\eta' + (\eta + 3\delta)/2} \frac{Y}{K} \frac{L}{EM} \cdot M \left(\frac{u^2 E^2 M^2}{L} \right)^{1/2 - \theta}$: the block's own trivial size $(Y/K)L/(EM)$ times $M(u^2 E^2 M^2/L)^{1/2 - \theta}$, the $\mathcal{K}_1$ -loss entering with the square root. If $uEM > L^{1/2}$ the block is bounded by its trivial size $(Y/K)L/(EM) < (Y/K)L^{1/2}u$, and these blocks sum geometrically. Since the block bound decays like $E^{-2\theta}$ against a trivial size like $1/E$, all dyadic E sum without truncation, and $M \leq Y^\eta$ gives $\sum_M M^{1-2\theta} \ll Y^{\eta(1-2\theta)}$. Hence, for the box,

$$\sum_{\text{blocks}} \ll H^{o(1)} Y^{C\eta' + (\eta + 3\delta)/2 + \eta(1-2\theta)} \frac{Y}{K} \left(L^{1/2 + \theta} u^{1-2\theta} + L^{1/2} u \right).$$

Step S8: the loss budget, the parameters and the summation

Sum over the boxes: for fixed K the bound is increasing in L , so the $O(\log H)$ values of L cost a logarithm and the worst box has $L \asymp Y$; the $O(\log H)$ values of $K \geq Y^{1-\delta}$ give $\sum_K Y/K \ll Y^\delta \log H$ — this factor Y^δ is a genuine loss of the reduction to boxes, the small moduli carrying a trivial size $Y^{1+\delta}$ in total. With $u^2/Y = u^3/H$ and the errors of (2), each piece therefore satisfies

$$\mathcal{P}_u(H/u) \ll \frac{H}{u} \left[\left(\frac{u^3}{H} \right)^{(1-2\theta)/2} H^\Lambda + H^{-\min(\delta, \eta, 2\eta' - \delta) + \varepsilon} \right], \quad \Lambda = C\eta' + \frac{\eta + 3\delta}{2} + \eta(1 - 2\theta) + \delta,$$

where the four terms of Λ are the kernel theorem's smoothness loss, the $\mathcal{K}_1$ -loss, the M -sum and the box reduction. Take $\delta = \eta = \eta' = t$ and $C = 6$: then $\Lambda = (10 - 2\theta)t$ and $\min(\delta, \eta, 2\eta' - \delta) = t$. For $u \leq H^{1/3-c}$ the first term is $\leq (H/u)H^{-3c(1-2\theta)/2+\Lambda}$, and summing over u with $\sum_{u \leq U} |w(u)| \ll U$ gives $\sum_u |w(u)|(H/u)(u^3/H)^{(1-2\theta)/2} \ll H^{1-3c(1-2\theta)/2+o(1)}$ for the main term and $\sum_{u \leq H^{1/3}} |w(u)|(H/u)^{1-t} \ll H^{1-t} \cdot H^{t/3}(\log H)^{O(1)} = H^{1-2t/3+o(1)}$ for the error. Balancing $\frac{3}{2}(1 - 2\theta)c - (10 - 2\theta)t = \frac{2}{3}t$ gives $t = \frac{9(1-2\theta)}{2(32-6\theta)}c$ and $\delta_S(c) = \frac{2}{3}t = \frac{3(1-2\theta)}{32-6\theta}c$; at $\theta = 7/64$ this is $t \approx 0.112c$ and $\delta_S(c) \approx 0.0748c$. This is Theorem 8. $\square$

Remark 9 (on the constant). An earlier version of this proof, and the working notes it was written from, stated $\delta_S(c) \approx c/12$; the budget there omitted the box-reduction loss Y^δ and did not halve the $\mathcal{K}_1$ -loss. The corrected budget is displayed above so that every loss can be checked against its step. Nothing in the shape of the theorem depends on the constant, and the constant itself is far from optimal (the parameters need not be equal, and $C = 6$ is a citation, not a computation).

Remark 10 (verification record). Theorem 5 was read three times by independent model readers (the third confirming the merged text) and Theorem 8's components in seven rounds, the last two on the text of this section, each recorded with its corrections in the repository (`research/explore/PROOFS-uniform.md`, §28–§48); the local count of Lemma 6 and the identities of Steps S5–S6 were brute-forced on small cases, including non-fundamental discriminants. The identities of part I on which the pieces rest are formally verified in Lean 4 (part I, *Data and code*). No human expert has checked this paper.

6 What remains: the band at $u = H^{1/3}$

Theorems 5 and 8 leave a neighbourhood of $\alpha = 1/3$ uncovered, and the two mechanisms fail there for unrelated reasons. On the upper side the balanced Type II atoms sit at $d^{1/4}$, where the available energy input for reciprocals is exactly at its threshold: what would be needed is a bound for the six-fold energy $J_6(N)$ below N^4 at $N \asymp m^{1/4}$, which is not known even for prime moduli. On the lower side the kernel method's diagonal, the number of orbits of the family, is $\asymp u$, and at $u \asymp Y^{1/2}$ it meets the trivial bound.

Three remarks fix what such a band costs, and what closing it would buy.

Remark 11 (a band is cheap, and a failing method is free). Inside any range of α one still has part III's Theorem E piece by piece, $\mathcal{P}_u(H/u) \ll (H/u)(\log H)^{-c} \log \log H$ uniformly in $u \leq H^{1-\varepsilon}$. Hence a band $[H^{\alpha_0-\delta}, H^{\alpha_0+\delta}]$ on which nothing better is available costs $\ll_f \delta \cdot H(\log H)^{1-c} \log \log H$, not $\delta H \log H$: a mechanism that degenerates can never make the total worse than what is already proved, and any band whose width tends to 0 improves Theorem E by that factor.

Remark 12 (a point gap costs $O(H)$). Suppose the savings on the two sides vanish only at $\alpha = 1/3$, say like $H^{-c|\alpha-1/3|}$, with implied constants that are powers of $\log H$. Summing over

dyadic blocks, $\sum_k H e^{-c|k \log 2 - \frac{1}{3} \log H|} = H(1 + 2/(e^{c \log 2} - 1)) = O(H/c)$: the blocks near the point contribute $O(H)$ in total. Since the main term $c_{\text{off}}(f)H$ comes from $u > H^{1/2+\varepsilon}$ and the remaining blocks are error terms, this would give $\text{Off}_f^*(H) = O(H)$, which is Hypothesis (E) of part I [5] and yields $\sum_{h \leq H} (1 - h/H)(S_f(h) - C(f)^2) = -\frac{1}{2}C(f) \log H + O_f(1)$: a bounded second-order term.

Remark 13 (what A_f needs). The constant of Conjecture 1 of part I requires $\text{Off}_f^*(H) = c_{\text{off}}(f)H + o(H)$, hence a saving factor $\gg \log H$ uniformly on $|\alpha - 1/3| \ll \log \log H / \log H$; a fixed positive power saving at $\alpha = 1/3$ itself suffices. A linearly vanishing saving, however small the rate, gives $O(H)$ but never $o(H)$.

Remark 14 (what is actually binding). An audit of every implied constant in the chain (research/explore/PROOFS-uniform.md §36) shows that no divisor-size loss constrains the argument: the divisor factors all occur under sums and average to powers of $\log H$, the kernel theorem's $\delta^{-O(1)}$ is δ^{-6} for the theorem used (Remark 7), its $(AD)^{o(1)}$ is $(\log AD)^{O(1)}$, and its $q^{o(1)}$ is 1 in the present application. What binds is the constant in Bourgain and Garaev's Hölder step, which grows as $k_1 \asymp 2/(9|\alpha - 1/3|) \rightarrow \infty$: the band on which the estimate is effective has width $\asymp (\log \log H / \log H)^{1/3}$. With the in-band fallback on part III's Theorem E this gives, once the map is written out, $\text{Off}_f^*(H) \ll H(\log H)^{0.63+o(1)}$ for $D < 0$, while everything outside the band sums to $O(H)$. The step from there to $\text{Off}_f^*(H) = O(H)$ therefore needs one thing only: an estimate that survives at the exceptional sizes $N \asymp m^{1/4}$ — an energy bound J_6 below N^4 for composite moduli, or a different tool there. A level-averaged spectral bound for the twist $e(c\bar{d}/x)$ with $x \asymp Y^{1/2}$ would serve the same purpose; neither is in the literature.

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